

Let, $p \in C$. $\therefore p \in A \cap p \in C$ (By defn. of AND). $p \in A \cap C$ (Defn. 2)
 $\therefore p \in B \cap C$ (By Defn. 3 from Section 2.1)
 $p \in B \cap p \in C$ (Defn. 2) $\Rightarrow p \in B$ (By Defn. of AND)
 By Defn. 3 from Section 2.1, $\boxed{A \subseteq B}$.

Let, p be an arbitrary element of B , i.e. $p \in B$.

Let, $p \notin C$. By Modus ponens, $p \in A \cap p \in B$. By Defn. of AND, $p \in A$.

Let, $p \in C$. $\therefore p \in B \cap p \in C$ (By defn. of AND). $p \in B \cap C$ (Defn. 2)

$\therefore p \in A \cap C$ (By Defn. 3 from Section 2.1)

$p \in A \cap p \in C$ (Defn. 2) $\Rightarrow p \in A$ (By Defn. of AND)

By Defn. 3 from Section 2.1, $\boxed{B \subseteq A}$

$\therefore \boxed{A=B}$ $\square$

39) Let A and B be subsets of a universal set U . Show that $A \subseteq B$ if and only if $\overline{B} \subseteq \overline{A}$

Solution: Given, $A \subseteq U$ and $B \subseteq U$.

Let, $A \subseteq B$. By Defn. 3 from Sec-2.1, $\forall x(x \in A \rightarrow x \in B)$. By universal instantiation, $(p \in A \rightarrow p \in B)$. By equivalence of contrapositive, $\neg(p \in B) \rightarrow \neg(p \in A)$. By Defn. of not belongs to, $p \notin B \rightarrow p \notin A$. By Defn. 5, $p \in \overline{B} \rightarrow p \in \overline{A}$. By Universal generalization, $\forall x(x \in \overline{B} \rightarrow x \in \overline{A})$. By Defn. 3 from Sec-2.1, $\overline{B} \subseteq \overline{A}$.

Let, $\overline{B} \subseteq \overline{A}$. By Defn. 3 from Sec-2.1, $\forall x(x \in \overline{B} \rightarrow x \in \overline{A})$. By universal instantiation, $(p \in \overline{B} \rightarrow p \in \overline{A})$. By equivalence of contrapositive, $\neg(p \in \overline{A}) \rightarrow \neg(p \in \overline{B})$. By Defn. 5, $\neg(p \notin A) \rightarrow \neg(p \notin B)$. By Defn. of not belongs to, $\neg(\neg(p \in A)) \rightarrow \neg(\neg(p \in B))$. By Double-negation Law of propositions, $p \in A \rightarrow p \in B$. By universal generalization, $\forall x(x \in A \rightarrow x \in B)$.

By Defn. 3 from Sec-2.1, $A \subseteq B$. $\square$

34) Let A, B and C be sets. Use the identity $A - B = A \cap \overline{B}$, which holds for any sets A and B , and the identities from Table 1 to show that $(A - B) \cap (B - C) \cap (A - C) = \emptyset$.